

Table I. Typical Engine Faults and Their Acoustic Characteristics

Fault Type	Description	Acoustic Pattern
Normal Operation	Smooth engine running	Stable and consistent sound
Misfire	Incomplete combustion	Irregular bursts, uneven rhythm
Knocking	Premature combustion	Sharp metallic knocking sound
Belt / Accessory Fault	Loose or worn belt	Squealing or chirping sound
Mechanical Fault	Internal component wear	Repetitive ticking or knocking

Table II. CNN Architecture Details

Layer	Type	Output Size	Details
1	Conv2D + ReLU	126×126×32	Kernel size: 3×3
2	MaxPooling	63×63×32	Pool size: 2×2
3	Conv2D + ReLU	61×61×64	Kernel size: 3×3
4	MaxPooling	30×30×64	Pool size: 2×2
5	Flatten	-	-
6	Dense	128	ReLU activation
7	Dropout	-	Rate: 0.3
8	Output	5	Softmax activation

The CNN architecture consists of multiple layers as detailed in Table II

TABLE III. Classification Performance of CNN Model

Class	Precision	Recall	F1-score	Support
Normal Engine	1.00	0.92	0.96	556
Combustion Fault	0.92	0.92	0.92	26
Mechanical Fault	0.69	0.98	0.81	180

Belt / Accessory Fault	1.00	0.91	0.95	11
Background Noise	0.90	0.62	0.73	91

TABLE IV. Comparison with Baseline Methods

Model	Accuracy	F1-score
SVM + MFCC (baseline)	78%	0.76
Proposed CNN + Mel- Spectrogram	86%	0.84